

Name: _____

In-Class Activity

PS200C: Causal Inference in the Social Sciences
Spring 2026

1. What is the causal question?

Which of the following best captures the causal question in the potential outcomes framework?

- (a) Compare average outcomes between treated and untreated groups
- (b) Compare outcomes for the same unit under treatment vs. control
- (c) Compare treated units to similar control units
- (d) Compare before-and-after outcomes for treated units

2. Observability

For a given unit i , classify each as *always observable*, *sometimes observable*, or *never observable*:

- $Y_i(1)$ and $Y_i(0)$ (both at once)
- D_i
- Y_i
- $\tau_i = Y_i(1) - Y_i(0)$

3. Difference in Means

Consider:

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]$$

Which is correct?

- (a) Always equals ATE
- (b) Always equals ATT
- (c) Equals ATT plus selection bias
- (d) Equals ATC

4. Direction of Bias

Suppose:

- $Y_i(1) = Y_i(0)$ for all i (no treatment effect)
- Treatment is more likely for units with higher $Y_i(1)$

What do you expect for:

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] ?$$

- (a) Zero

- (b) Positive
- (c) Negative

5. **Science Table**

i	D_i	$Y_i(1)$	$Y_i(0)$
1	1	10	4
2	1	1	2
3	0	3	3
4	0	5	2

Compute:

- ATE:
- ATT:

6. **Bonus:**

Why does

$$\mathbb{E}[Y_i(1) \mid D_i = 1] = \mathbb{E}[Y_i \mid D_i = 1]$$

hold?